

LIST OF PUBLICATIONS

All papers submitted to the arxiv can be obtained at http://arxiv.org/a/schweigert_c_1

Publications in refereed journals:

1. “*On the classification of $N = 2$ superconformal coset models*”
Commun. Math. Phys. 149 (1992) 425 - 431
2. “*Non-Hermitian symmetric $N = 2$ coset models, Poincaré polynomials and string compactification*”
(with J. Fuchs)
Nucl. Phys. B 411 (1994) 181 - 222, arXiv:hep-th/9304133v1
3. “*Level-rank duality of WZW theories and isomorphisms of $N = 2$ coset models*,”
(with J. Fuchs)
Annals of Physics 234 (1994) 102 - 140, arXiv:hep-th/9307107v2
4. “*Poincaré polynomials and level-rank dualities in the $N = 2$ coset construction*,”
Theor. Math. Phys. 98 (1994) 326 - 334, arXiv:hep-th/9311168v1
5. “*On the configuration space of gauge theories*,”
(with J. Fuchs and M.G. Schmidt)
Nucl. Phys. B 426 (1994) 107 - 128, hep-th/9404059
6. “*Modular invariants and fusion rule automorphisms from Galois theory*,”
(with J. Fuchs, B. Gato-Rivera and A.N. Schellekens)
Phys. Lett. B 334 (1994) 113 - 120, arXiv:hep-th/9405153v1
7. “*Extended geometry of black holes*,”
(with K. Peeters and J.W. van Holten)
Class. Quantum Grav. 12 (1995) 173 - 179, arXiv:gr-qc/9407006v1
8. “*Galois modular invariants of WZW models*,”
(with J. Fuchs and A.N. Schellekens)
Nucl. Phys. B 437 (1995) 667 - 694, arXiv:hep-th/9410010v1
9. “*Quasi-Galois symmetries of the modular S -matrix*,”
(with J. Fuchs and A.N. Schellekens)
Commun. Math. Phys. 176 (1996) 447 - 465, arXiv:hep-th/9412009v1
10. “*On the extended Poincaré polynomial*,”
(with M. Kreuzer)
Phys. Lett. B 352 (1995) 276 - 285, arXiv:hep-th/9503174v1
11. “*From Dynkin diagram symmetries to fixed point structures*,”
(with J. Fuchs and A.N. Schellekens)
Commun. Math. Phys. 180 (1996) 39 - 97, arXiv:hep-th/9506135v3

12. “*The resolution of field identification fixed points in diagonal coset theories*,”
(with J. Fuchs and A.N. Schellekens)
Nucl. Phys. B 461 (1996) 371 - 404, arXiv:hep-th/9509105v1
13. “*A matrix S for all simple current extensions*,”
(with J. Fuchs and A.N. Schellekens)
Nucl. Phys. B 473 (1996) 323 - 366, arXiv:hep-th/9601078v2
14. “*Some automorphisms of Generalized Kac-Moody algebras*,”
(with J. Fuchs and U. Ray)
Journal of Algebra 191 (1997) 518 - 540, arXiv:q-alg/9605046v1
15. “*WZW fusion rings in the limit of infinite level*,”
(with J. Fuchs)
Commun. Math. Phys. 185 (1997) 641 - 670, arXiv:hep-th/9609124v1
16. “*On moduli spaces of flat connections with non-simply connected structure group*,”
Nucl. Phys. B 492 (1997) 743 - 755, arXiv:hep-th/9611092v1
17. “*Systematic approach to cyclic orbifolds*,”
(with L. Borisov and M.B. Halpern)
Int. J. Mod. Phys. A 13 (1998) 125 - 168, arXiv:hep-th/9811211v1
18. “*A classifying algebra for boundary conditions*,”
(with J. Fuchs)
Phys. Lett. B 414 (1997) 251 - 259, arXiv:hep-th/9708141v1
19. “*A note on the geometry of CHL heterotic strings*,”
(with W. Lerche, R. Minasian, and S. Theisen)
Phys. Lett. B 424 (1998) 53 - 59, arXiv:hep-th/9711104v1
20. “*Branes: from free fields to general backgrounds*,”
(with J. Fuchs)
Nucl. Phys. B 530 (1998) 99 - 136, arXiv:hep-th/9712257v2
21. “*The action of outer automorphisms on bundles of chiral blocks*,”
(with J. Fuchs)
Commun. Math. Phys. 206 (1999) 691 - 736. arXiv:hep-th/9805026v3
22. “*Completeness of boundary conditions for the critical three-state Potts model*,”
(with J. Fuchs)
Phys. Lett. B. 441 (1998) 141 - 146, arXiv:hep-th/9806121v1
23. “*Orbifold analysis of broken bulk symmetries*,”
(with J. Fuchs)
Phys. Lett. B. 447 (1999) 266 - 276, arXiv:hep-th/9811211v1
24. “*Symmetry breaking boundaries: I. General theory*,”
(with J. Fuchs)
Nucl. Phys. B, 558 (1999) 419 - 483, arXiv:hep-th/9902132v2
25. “*Symmetry breaking boundary conditions and WZW orbifolds*,”
(with L. Birke and J. Fuchs)
Adv. Th. Math. Phys. 3 (1999) 671 - 726, arXiv:hep-th/9905038v1

26. “*Symmetry breaking boundaries: II. More structures, examples,*”
(with J. Fuchs)
Nucl. Phys. B 568 (2000) 543 - 593, arXiv:hep-th/9908025v1
27. “*New maverick coset theories,*”
(with B. Pedrini and J. Walcher)
Phys. Lett. B 466 (1999) 206 - 210, arXiv:hep-th/9908185v1
28. “*The geometry of WZW branes,*”
(with G. Felder, J. Fröhlich and J. Fuchs)
J. Geom. Phys. 34 (2000) 162 - 190, arXiv:hep-th/9909030v3
29. “*Conformal boundary conditions and three-dimensional topological field theory,*”
(with G. Felder, J. Fröhlich and J. Fuchs)
Phys. Rev. Lett. 84 (2000) 1659 - 1662, arXiv:hep-th/9909140v2
30. “*Correlation functions and boundary conditions in RCFT and three-dimensional topology,*”
(with G. Felder, J. Fröhlich and J. Fuchs)
Compositio Math. 131 (2002) 189 - 237, arXiv:hep-th/9912239v2
31. “*Universality in quantum Hall systems: coset construction of incompressible states,*”
(with J. Fröhlich, B. Pedrini and J. Walcher)
J. Stat. Phys 103 (2001) 527 - 567. Dedicated to the memory of Quin Luttinger.
arXiv:cond-mat/0002330v2 [cond-mat.mes-hall]
32. “*Flux Stabilization of D-branes,*”
(with C. Bachas and M. Douglas)
JHEP 5 (2000) Paper 048, 18 pp. arXiv:hep-th/0003037v2
33. “*Projections in string theory and boundary states for Gepner models,*”
(with J. Fuchs and J. Walcher)
Nucl. Phys. B 588 (2000) 110 - 148, arXiv:hep-th/0003298v1
34. “*Solitonic sectors, alpha induction and symmetry breaking boundaries,*”
(with J. Fuchs)
Phys. Lett. B 490 (2000) 163 - 172, arXiv:hep-th/0006181v2
35. “*Boundaries, crosscaps and simple currents,*”
(with J. Fuchs, L.R. Huiszoon, A.N. Schellekens and J. Walcher)
Phys. Lett. B 495 (2000) 427 - 434, arXiv:hep-th/0007174v2
36. “*Boundary Fixed Points, Enhanced Gauge Symmetry and Singular Bundles on K3,*”
(with J. Fuchs, P. Kaste, W. Lerche, A. Lütken and J. Walcher)
Nucl. Phys. B 598 (2001) 57 - 72, arXiv:hep-th/0007145v1
37. “*Flux stabilization in compact groups,*”
(with P. Bordalo and S. Ribault)
JHEP 10 (2001) Paper 036, 16 pp. arXiv:hep-th/0108201v2

38. “*D-Branes on ALE spaces and the ADE classification of conformal field theories*,”
(with W. Lerche and A. Lütken)
Nucl. Phys. B 622 (2002) 269 - 278, arXiv:hep-th/0006247v3
39. “*Conformal Correlation Functions, Frobenius Algebras and Triangulations*,”
(with J. Fuchs and I. Runkel)
Nucl. Phys. B 624 (2002) 452 - 468, arXiv:hep-th/0110133v2
40. “*A reason for fusion rules to be even*,”
(with J. Fuchs and I. Runkel)
J. of Physics A 35 (2002) L255-260. arXiv:math/0110257v1 [math.QA]
41. “*TFT construction of RCFT correlators I: Partition functions*,”
(with J. Fuchs and I. Runkel)
Nucl. Phys. B 646 (2002) 353-544. arXiv:hep-th/0204148v2
42. “*An algorithm for twisted fusion rules*,”
(with T. Quella and I. Runkel)
Adv. Th. Math. Phys. 6 (2002) No. 2, 197- 206. arXiv:math/0203133v2 [math.QA]
43. “*The world sheet revisited*,”
(with J. Fuchs)
Fields Institute Comm. 39 (2003) 241-249. arXiv:hep-th/0105266v1
44. “*Category theory for conformal boundary conditions*,”
(with J. Fuchs)
Fields Institute Comm. 39 (2003) 25-70. arXiv:math/0106050v4 [math.CT]
45. “*Topologization of chiral representations*,”
(with F. Conrady)
Commun. Math. Phys. 245 (2004) 429-448. arXiv:hep-th/0210215v2
46. “*TFT construction of RCFT correlators II: Unoriented world sheets*,”
(with J. Fuchs and I. Runkel)
Nucl. Phys. B. 678 (2004) 511-637. arXiv:hep-th/0306164v2
47. “*Kramers-Wannier duality from conformal defects*,”
(with J. Fröhlich, J. Fuchs and I. Runkel)
Phys. Rev. Lett. 93 (2004) 070601, arXiv:cond-mat/0404051v2 [cond-mat.stat-mech]
48. “*TFT construction of RCFT correlators III: Simple currents*,”
(with J. Fuchs and I. Runkel)
Nucl. Phys. B. 694 (2004) 277-353, arXiv:hep-th/0403157v2
49. “*Correspondences of ribbon categories*,”
(with J. Fröhlich, J. Fuchs and I. Runkel)
Adv. Math. 199 (2006) 192-329, arXiv:math/0309465v3 [math.CT]
50. “*TFT construction of RCFT correlators IV: Structure constants and correlation functions*,”
(with J. Fuchs and I. Runkel)
Nucl. Phys. B 715 (2005) 539-638, arXiv:hep-th/0412290v2

51. “*TFT construction of RCFT correlators V: Proof of modular invariance and factorisation,*”
(with J. Fjelstad, J. Fuchs and I. Runkel)
Theory and Applications of Categories 16 (2006) 342-433, arXiv:hep-th/0503194v2
52. “*Duality and defects in rational conformal field theory,*”
(with J. Fröhlich, J. Fuchs and I. Runkel)
Nucl. Phys. B 763 (2007) 354-430, arXiv:hep-th/0607247v2
53. “*Unoriented WZW Models and Holonomy of Bundle Gerbes,*”
(with U. Schreiber and K. Waldorf)
Commun. Math. Phys. 274 (2007) 31-64, arXiv:hep-th/0512283v2
54. “*Topological defects for the free boson CFT,*”
(with J. Fuchs, M. Gaberdiel and I. Runkel)
J. Phys. A: Math. Theor. 40 (2007) 11403-11440, arXiv:0705.3129v1 [hep-th]
55. “*Ribbon categories and (unoriented) CFT: Frobenius algebras, automorphisms, reversions,*”
(with J. Fuchs and I. Runkel)
Contemporary Mathematics 431 (2007) 203-224, arXiv:math/0511590v1 [math.CT]
56. “*Topological and conformal field theory as Frobenius algebras,*”
(with J. Fjelstad, J. Fuchs and I. Runkel)
Contemporary Mathematics 431 (2007) 225-247, arXiv:math/0512076v2 [math.CT]
57. “*Bi-branes: Target Space Geometry for World Sheet topological Defects,*”
(with J. Fuchs and K. Waldorf)
J. Geom. Phys. 58 (2008) 576-598, arXiv:hep-th/0703145v1
58. “*Uniqueness of open/closed rational CFT with given algebra of open states,*”
(with J. Fjelstad, J. Fuchs and I. Runkel)
Adv. Theor. Math. Phys. 12 (2008) 1281 - 1373, arXiv:hep-th/0612306v2
59. “*The fusion algebra of bimodule categories,*”
(with J. Fuchs and I. Runkel)
Applied Categ. Structures 16 (2008) 123 - 140, arXiv:math/0701223v1 [math.CT]
60. “*Kramers-Wannier dualities for WZW theories and minimal models,*”
(with E. Tsouchnika)
Commun. Contemp. Math 10 (2008) 773-789, arXiv:0710.0783v2 [hep-th]
61. “*Some remarks on defects and T-duality,*”
(with G. Sarkissian)
Nucl. Phys. B 819 (2009) 478-490, arXiv:0810.3159v2 [hep-th]
62. “*The three-dimensional origin of the classifying algebra,*”
(with J. Fuchs and C. Stigner)
Nucl. Phys. B 824 (2010) 333-364, arXiv:0907.0685v1 [hep-th]
63. “*Twenty five years of two-dimensional rational conformal field theory,*”
(with J. Fuchs and I. Runkel)
J. Math. Phys. 51 (2010) 015210, arXiv:0910.3145v2 [hep-th]

64. “*On the Rosenberg-Zelinsky sequence in abelian monoidal categories,*”
(with T. Barmeier, J. Fuchs and I. Runkel)
J. reine angew. Math. (Crelle’s Journal) 642 (2010) 1-36, arXiv:0801.0157
65. “*Module categories for permutation modular invariants,*”
(with T. Barmeier, J. Fuchs and I. Runkel)
Int. Math. Research Notices 16 (2010) 3067-3100, arXiv:0812.0986v2 [math.CT]
66. “*Hopf algebras and finite tensor categories in conformal field theory,*”
(with J. Fuchs)
Revista de la Unión Matemática Argentina 51 (2010) 43-90, arXiv:1004.3405 [hep-th]
67. “*Equivariance in higher geometry,*”
(with T. Nikolaus)
Adv. Math. 226 (2011) 3367-3408, arXiv:1004.4558v1 [math.AT]
68. “*The classifying algebra for defects,*”
(with J. Fuchs and Carl Stigner)
Nucl. Phys. B 843 (2011) 673-723, arXiv:1007.0401v1 [hep-th]
69. “*Modular categories from finite crossed modules*”
(with J. Maier)
Journal of Pure and Applied Algebra, 215 (2011) 2196-2208, arXiv:1003.2070 [math.QA]
70. “*A Geometric Construction for Permutation Equivariant Categories from Modular Functors,*”
(with T. Barmeier)
Transform. Groups 16 (2011) 287-337, arXiv:1004.1825v1 [math.QA]
71. “*Equivariant modular categories via Dijkgraaf-Witten theory,*”
(with J. Maier and Th. Nikolaus)
Adv. Theor. Math. Phys. 16 (2012) 289-358, arXiv:1103.2963v1 [math.QA]
72. “*Modular invariant Frobenius algebras from ribbon Hopf algebra automorphisms,*”
(with J. Fuchs and C. Stigner)
J. Algebra 363 (2012) 29-72, arXiv:1106.0210v1 [math.QA]
73. “*Strictification of weakly equivariant Hopf algebras,*”
(with J. Maier and Th. Nikolaus)
Bull. Belgian Math. Soc. 20 (2013), 269-286, arXiv:1109.0236v1 [math.RA]
74. “*Bicategories for boundary conditions and for surface defects in 3-d TFT,*”
(with J. Fuchs and A. Valentino)
Commun. Math. Phys. 321 (2013) 543-575, arXiv:1203.4568v2 [hep-th]
75. “*Higher genus mapping class group invariants from factorizable Hopf algebras,*”
(with J. Fuchs and C. Stigner)
Adv. Math. 250 (2014) 285-319, arXiv:1207.6863v1 [math.QA]
76. “*From non-semisimple Hopf algebras to correlation functions for logarithmic CFT,*”
(with J. Fuchs and C. Stigner)
J. Phys. A: Math. Theor. 46 (2013) 494008.

77. “*A geometric approach to boundaries and surface defects in Dijkgraaf-Witten theories,*”
(with J. Fuchs and A. Valentino)
Commun. Math. Phys. 332 (2014) 981-1015, arXiv:1307.3632 [hep-th]
78. “*A note on permutation twist defects in topological bilayer phases,*”
(with J. Fuchs)
Lett. Math. Phys. 104 (2014) 1385-1405, arXiv:1310.1329 [hep-th]
79. “*A Serre-Swan theorem for gerbe modules on étale Lie groupoids,*”
(with C. Tropp and A. Valentino)
Theory Appl. Cat. 29 (2014) 819–835, arXiv:1401.2824 [math.AT]
80. “*On the Brauer groups of symmetries of abelian Dijkgraaf-Witten theories,*”
(with J. Fuchs, J. Priel and A. Valentino)
Commun. Math. Phys. 339 (2015) 385-405, arXiv:1404.6646 [hep-th]
81. “*Partially dualized Hopf algebras have equivalent Yetter-Drinfel’d modules,*”
(with A. Barvels and S. Lentner)
J. Algebra 430 (2015) 303-342, arXiv:1402.2214 [math.QA]
82. “*A trace for bimodule categories,*”
(with J. Fuchs and G. Schaumann)
Applied Categorical Structures 25 (2017) 227-268, arXiv:1412.6968 [math.CT]
83. “*Consistent systems of correlators in non-semisimple conformal field theory,*”
(with J. Fuchs)
Adv. Math. 307 (2017) 598-639, arXiv:1604.01143 [math.QA]
84. “*Frobenius algebras and homotopy fixed points of group actions on bicategories,*”
(with J. Hesse and A. Valentino)
Theory and Applications of Categories 32 (2017). 652-681, arXiv:1607.05148 [math.QA]
85. “*A GNS construction of three-dimensional abelian Dijkgraaf-Witten theories,*”
(with L. Müller)
Rev. Math. Phys. 30 (2018) 1850005, arXiv:1703.05018 [math.QA]
86. “*Hochschild Cohomology and the Modular Group,*”
(with S. Lentner, S. Mierach and Y. Sommerhäuser)
J. Algebra 507 (2018) 400-420, arXiv:1707.04032 [math.RA]
87. “*The logarithmic Cardy case: Boundary states and annuli,*”
(with J. Fuchs, T. Gannon and G. Schaumann)
Nucl. Phys. B 930 (2018) 287-327, arXiv:1712.01922 [math.QA]
88. “*A Parallel Section Functor for 2-Vector Bundles,*”
(with L. Woike)
Theory and Applications of Categories 33 (2018) 644-690, arXiv:1711.08639 [math.CT]
89. “*On unrolled Hopf algebras,*”
(with N. Andruskiewitsch)
J. Knot Theory and its Ramif. 27 (2018) 1850053, arXiv:1701.00153 [math.QA]

90. “*Orbifold Construction for Topological Field Theories*,”
(with L. Woike)
J. Pure and Applied Algebra (223) 2019 1167-1192, arXiv:1705.05171 [math.QA]
91. “*A Parallel Section Functor for 2-Vector Bundles*,”
(with L. Woike)
Theory and Applications of Categories 33 (2018) 644-690, arXiv:1711.08639 [math.CT]
92. “*Eilenberg-Watts calculus for finite categories and a bimodule Radford S^4 theorem*,”
(with J. Fuchs and G. Schaumann)
Trans. Amer. Math. Soc. 373 (2020) 1-40, arXiv:1612.04561 [math.RT]
93. “*Extended Homotopy Quantum Field Theories and their Orbifoldization*,”
(with L. Woike)
Journal of Pure and Applied Algebra (224) 2020, 106213 arXiv:1802.08512 [math.QA]
94. “*On isotypic decompositions for non-semisimple Hopf algebras*”
(with V. Koppen and E. Meir)
Algebr. Represent. Theor. (2021). <https://doi.org/10.1007/s10468-021-10029-x>, arXiv:1910.13161 [math.QA]
95. “*Matrix product operator symmetries and intertwiners in string-nets with domain walls*,”
(with L. Lootens, J. Fuchs, J. Haegeman and F. Verstraete)
SciPost Phys. 10, 053 (2021), arXiv:2008.11187 [quant-ph]
96. “*Bulk from boundary in finite CFT by means of pivotal module categories*,”
(with J. Fuchs)
Nucl. Phys. B 967 (2021) 115392, arXiv:2012.10159 [hep-th]
97. “*CFT correlators for Cardy bulk fields via string-net models*,”
(with Y. Yang)
SIGMA 17 (2021) 040, arXiv:1911.10147 [hep-th]
98. “*Homotopy Coherent Mapping Class Group Actions and Excision for Hochschild Complexes of Modular Categories*,”
(with L. Woike)
Adv. Math. 386(2021)107814, arXiv:2004.14343 [math.QA]
99. “*Internal natural transformations and Frobenius algebras in the Drinfeld center*,”
(with J. Fuchs)
Transformation Groups 28 (2023) 733, arXiv:2008.04199 [math.CT]
100. “*The Hochschild Complex of a Finite Tensor Category*,”
(with L. Woike)
Algebraic & Geometric Topology 21 (2021) 3689–3734, arXiv:1910.00559 [math.QA]
101. “*A modular functor from state sums for finite tensor categories and their bimodules*,”
(with J. Fuchs and G. Schaumann)
Theory and Applications of Categories 38 (2022) 436-594, arXiv:1911.06214 [math.QA]
102. “*Frobenius-Schur Indicators and the Mapping Class Group of the Torus*,”
(with J. Farnsteiner)
Lett. Math. Phys. 112 (2022) 39, arXiv:2104.12742 [math.QA]

103. “*Equivariant Morita theory for graded tensor categories*,”
(with C. Galindo and D. Jaklitsch)
Bull. Belg. Math. Soc. Simon Stevin 29 (2022) 145–171, arXiv:2106.07440 [math.QA]
104. “*The Trace Field Theory of a Finite Tensor Category*,”
(with L. Woike)
Algebras and Representation Theory (2022), arXiv:2103.15772 [math.QA]
105. “*Domain walls between 3d phases of Reshetikhin-Turaev TQFTs*,”
(with V. Koppen, V. Mulevicius and I. Runkel)
Commun. Math. Phys. 396 (2022) 1187–1220, arXiv:2105.04613 [hep-th]
106. “*Davydov-Yetter cohomology, comonads and Ocneanu rigidity*,”
(with A. Gainutdinov and J. Haferkamp)
Adv. Math. 414 (2023), Paper No. 108853, arXiv:1910.06094 [math.QA]
107. “*The differential graded Verlinde Formula and the Deligne Conjecture*,”
(with L. Woike)
Proc. London Math. Soc. 126 (2023) 1811–1841, arXiv:2105.01596 [math.QA]
108. “*Homotopy Invariants of Braided Commutative Algebras and the Deligne Conjecture for Finite Tensor Categories*,”
(with L. Woike)
Adv. Math. 422 (2023) 109006, arXiv:2204.09018 [math.QA]
109. “*A G -equivariant String-Net Construction*,”
(with A. DeLazzer Meunier and M. Traube)
Annales H. Poincaré 25 (2024) 297–345, arXiv:2304.00106 [math.QA]
110. “*Spherical Morita contexts and relative Serre functors*,”
(with J. Fuchs, C. Galindo and D. Jaklitsch)
Kyoto J. Math. 65 (2025) 537–594, arXiv:2207.07031 [math.QA]
111. “*Davydov–Yetter cohomology and relative homological algebra*,”
(with M. Faitg and A. Gainutdinov)
Sel. Math. New Ser. 30, 26 (2024), arXiv:2202.12287 [math.QA]
112. “*Twisted Drinfeld Centers and Framed String-Nets*,”
(mit H. Knötzele und M. Traube)
Quantum Topol. 15 (2024) 537–566, arXiv:2302.14779 [math.QA]
113. “*The evaluation of graphs on surfaces for state-sum models with defects*,”
(with J. Farnsteiner)
SIGMA 20 (2024) 102, arXiv:2312.01946 [math.QA]
114. “*All product eigenstates in Heisenberg models from a graphical construction*,”
(with F. Gerken, I. Runkel and T. Posske)
Phys. Rev. Research 7 (2025) L012008 arXiv:2310.13158 [cond-mat.str-el]
115. “*Algebraic structures in two-dimensional conformal field theory*,”
(with J. Fuchs, S. Wood and Y. Yang)
Encyclopedia of Mathematical Physics (Second Edition). Elsevier Ltd, Volume 3. p. 604–617
arXiv:2305.02773

116. “*A manifestly Morita-invariant construction of Turaev-Viro invariant,*”
(with J. Fuchs, C. Galindo and D. Jaklitsch)
ZMP-HH/24-13, Hamburger Beiträge zur Mathematik Nr. 970, arXiv:2407.10018 [math.QA],
accepted for publication in Quantum topology.
117. “*Duality structures for module categories of vertex operator algebras and the Feigin Fuchs boson,*”
(with R. Allen, S. Lentner and S. Wood)
ZMP-HH/21-16, Hamburger Beiträge zur Mathematik Nr. 904, arXiv:2107.05718 [math.QA],
accepted for publication with selecta.
118. “*Grothendieck-Verdier duality in categories of bimodules and weak module functors,*”
(with J. Fuchs, G. Schaumann and S. Wood)
Contemporary Mathematics 813, 2025, arXiv:2306.17668 [math.CT]
119. “*String-net models for pivotal bicategories,*”
(with J. Fuchs and Y. Yang)
Theory and Applications of Categories 44 (2025) 474-543, arXiv:2302.01468 [math.QA]
120. “*The Lyubashenko Modular Functor for Drinfeld Centers via Non-Semisimple String-Nets,*”
(with L. Müller, L. Woike and Y. Yang)
Adv. Math. 488 (2026) 110770, arXiv:2312.14010 [math.QA]

Preprints:

1. “*Grothendieck-Verdier module categories, Frobenius algebras and relative Serre functors,*”
(with J. Fuchs, G. Schaumann and S. Wood)
ZMP-HH/24-11, Hamburger Beiträge zur Mathematik Nr. 966, arXiv:2405.20811 [math.CT]
2. “*Excision for Spaces of Admissible Skeins,*”
(with I. Runkel and Y.H. Tham)
ZMP-HH/24-14, Hamburger Beiträge zur Mathematik Nr. 971, arXiv:2407.09302 [math.QA]
3. “*An adjunction theorem for Davydov-Yetter cohomology and infinitesimal braidings,*”
(with M. Faitg and A. Gainutdinov)
ZMP-HH/24-26, Hamburger Beiträge zur Mathematik Nr. 981, arXiv:2411.19111 [math.QA]
4. “*Surface Diagrams for Frobenius Algebras and Frobenius-Schur Indicators in Grothendieck-Verdier Categories,*”
(with M. Demirdilek)
Accepted for publication in Higher Structures. arXiv:2503.13325 [math.CT]
5. “*A boundary characterization of Turaev-Viro TQFTs,*”
(with M.-N. Steffen)
ZMP-HH/25-22, Hamburger Beiträge zur Mathematik Nr. 1003, arXiv:2508.13759 [math.QA]
6. “*The 2-categorical S-matrix of a braided fusion 1-category is a character table,*”
(with A. Hofstetter)
ZMP-HH/26-17, Hamburger Beiträge zur Mathematik Nr. 1004, arXiv:2601.14168 [math.QA]
7. “*Deformations of mixed associators in module categories,*”
(with M. Faitg, A. Gainutdinov and J.-O. Willprecht)
ZMP-HH/26-20, Hamburger Beiträge zur Mathematik Nr. 1007, arXiv:2604.00837 [math.QA]
8. “*Linearly distributive coherence in the absence of units,*”

- (with M. Demirdilek and C. Reiher)
 ZMP-HH/26-23, Hamburger Beiträge zur Mathematik Nr. 1010, arXiv:2605.03113 [math.CO]
9. “*Modular functors and CFT correlators via double categories*,”
 (with J. Fuchs and Y. Yang)
 ZMP-HH/26-24, Hamburger Beiträge zur Mathematik Nr. 1011, arXiv:2605.03708 [math.QA]
10. “*Hochschild Cohomology, Modular Tensor Categories, and Mapping Class Groups II*,”
 (with S. Lentner, S. Mierach and Y. Sommerhäuser)
 ZMP-HH/26-25, Hamburger Beiträge zur Mathematik Nr. 1012, arXiv:2608.11193 [math.QA]

Books:

1. “*Symmetries, Lie Algebras and Representations*”,
 (with J. Fuchs)
 Cambridge Monographs on Mathematical Physics, Cambridge University Press, 1997. xii+438 pp.
 Paperback edition: October 2003, ISBN: 0521560012.
2. “*Softwarepraktikum - Analysis und Lineare Algebra*”,
 (with D. Bahns)
 Vieweg, 2007. ISBN 978-3-8348-0370-2
3. “*String-net construction of RCFT correlators*,”
 (with J. Fuchs and Y. Yang)
 Springer Briefs in Mathematical Physics 45 (2022), <https://doi.org/10.1007/978-3-031-14682-4>
 arXiv:2112.12708 [math.QA]
4. “*Hochschild Cohomology, Modular Tensor Categories, and Mapping Class Groups*,”
 (with S. Lentner, S. Mierach and Y. Sommerhäuser)
 Springer Briefs in Mathematical Physics 44 (2023), arXiv:2003.06527 [math.QA]
5. “*A First Course in Topological Field Theory*”
 (with J. Fuchs and L. Woike)
 AMS, University Lecture Series 80 (2026) ISBN: 978-1-4704-8507-8

Books edited:

1. “*Noncommutative structures in mathematics and Physics*”,
 (with S. Caenepeel, J. Fuchs, S. Gutt, A. Stolin and F. Van Oystaeyen)
 Royal Flemish Academy of Belgium, Brussels, 2010, ISBN 9789065690616.
2. “*Conformal Field Theories and Tensor Categories*”,
 (with C. Bai, J. Fuchs, Y. Huang, L. Kong and I. Runkel)
 Proceedings of a Workshop Held at Beijing International Center for Mathematical Research
 Springer Verlag Berlin, 2014.
3. “*Representation theory – Current Trends and Perspectives*”,
 (with H. Krause, P. Littellmann, G. Malle and K.-H. Neeb)
 EMS Series of Congress Reports, EMS Publishing House, 2017

Proceedings:

1. *"Twining characters, orbit Lie algebras, and fixed point resolution,"*
(with J. Fuchs and A.N. Schellekens)
in: New Trends in Quantum Field Theory; A.Ch. Ganchev, R. Kerner and I. Todorov eds.,
Heron Press Sofia, 1996. arXiv:q-alg/9511026v1
2. *"Fixed point resolution in conformal field theory,"*
(with J. Fuchs and A.N. Schellekens)
in: Group21, Physical Applications and Mathematical Aspects of Geometry, Groups, and
Algebras; H.-D. Doebner, W. Scherer, and C. Schulte eds., World Scientific, 1997.
arXiv:hep-th/9612093v1
3. *"Twining characters and orbit Lie algebras,"*
(with J. Fuchs, U. Ray and A.N. Schellekens)
in: Group21, Physical Applications and Mathematical Aspects of Geometry, Groups, and
Algebras; H.-D. Doebner, W. Scherer, and P. Nattermann eds., World Scientific, 1997.
arXiv:hep-th/9612060v1
4. *"Classifying algebras for boundary conditions and traces on spaces of conformal blocks,"*
(with J. Fuchs)
in: Theory of elementary particles; H. Dorn, D. Lüst, and G. Weigt eds.,
Wiley-VCH, 1998, p. 13-18, arXiv:hep-th/9801191v1
5. *"D-brane conformal field theory,"*
(with J. Fuchs)
in: Theory of elementary particles; H. Dorn, D. Lüst, and G. Weigt eds.,
Wiley-VCH, 1998, p. 7-12, arXiv:hep-th/9801190v1
6. *"Bundles of chiral blocks and boundary conditions in CFT,"*
(with J. Fuchs)
in: Lie Theory and its applications in physics III; H.-D. Doebner et al eds.,
World Scientific, 2000, p. 221-236. arXiv:hep-th/0001005v1
7. *"D-brane conformal field theory and bundles of conformal blocks,"*
(with J. Fuchs)
European Congress of Mathematics 2000; C. Casacuberta et al eds.,
Birkhäuser, Progress in mathematics 201, 2001, p. 489-499. arXiv:math/0004034v1 [math.QA]
8. *"Solitonic sectors, conformal boundary conditions and three-dimensional topological field theory,"*
(with J. Fuchs)
in: Non-perturbative quantum effects 2000; D. Bernard et al eds., JHEP Proceedings 2000.
arXiv:hep-th/0009111v2
9. *"Lie algebra automorphisms in conformal field theory,"*
(with J. Fuchs)
in: Recent Developments in Infinite-Dimensional Lie Algebras and Conformal Field Theory,
S. Berman et al eds., Contemporary Mathematics Volume 297, AMS, 2002, p. 117-142.
arXiv:math/0011160v1 [math.QA]

10. “*Conformal field theory, boundary conditions and applications to string theory*,”
(with J. Fuchs and J. Walcher)
in: Non-perturbative QFT methods and their applications; Z. Horvath and L. Palla eds.,
World Scientific, 2001. p.37-93. arXiv:hep-th/0011109v1
11. “*D-branes in group manifolds and flux stabilization*,”
in: Proceedings of the Ninth Marcel Grossmann Meeting on General Relativity;
V.G. Gurzadyan, R.T. Jantzen and R. Ruffini eds., World Scientific, 2002, p. 1217 - 1221.
arXiv:hep-th/0012045v1
12. “*Conformal boundary conditions and 3D topological field theory*.”
(with J. Fuchs and I. Runkel)
in: Statistical Field Theories; A. Cappelli and G. Mussardo eds., Kluwer Academic
Publishers, 2002, p. 185-194. arXiv:hep-th/0110158v1
13. “*Strings, symmetries and representations*.”
(with J. Fuchs and I. Runkel)
in: Group 24: Physical and mathematical aspects of symmetries. Institute of Physics
Conference Series Nr. 173, IOP Publishing 2003, p. 153-160.
14. “*Boundaries, defects and Frobenius algebras*.”
(with J. Fuchs and I. Runkel)
Ann. Henri Poincaré 4, Suppl. 1 (2003) S175-182 and Fortschritte der Physik, 51 (2003)
850-855, arXiv:hep-th/0302200v1
15. “*Open Strings and 3d Topological Field Theory*.”
(with J. Fuchs and I. Runkel)
Proceedings des ASI Summer Institute, Cargèse, Kluwer Academic, 2003, p. 461-464
16. “*Algebras in tensor categories and coset conformal field theories*,”
(with J. Fröhlich, J. Fuchs and I. Runkel)
Fortschritte der Physik, 52 (2004) p. 672-677, arXiv:hep-th/0309269v1
17. “*Lie algebras, Fuchsian differential equations and CFT correlation functions*,”
(with J. Fuchs and I. Runkel)
in: Ramanujan International Symposium on Kac-Moody Lie Algebras and Applications;
N. Sthanumoorthy and K. Misra eds., Contemporary Mathematics Volume 343,
AMS 2004, p. 115-129. arXiv:hep-th/0301181v1
18. *7 contributions*
in the Concise encyclopedia of supersymmetry and noncommutative structures in
mathematics and physics. St. Duplij, W. Siegel and J. Bagger editors. Kluwer, 2004.
Boundary state (p. 67), Coxeter number (p. 118), Dynkin index (p. 133) Groupoid (p. 179),
Sugawara construction (p. 384), p. 492 Twisted sector (p. 492), Vertex operator
algebra (p. 499). Contribution 1,2,3 and 7 with Jürgen Fuchs.
19. “*Picard groups in rational conformal field theory*,”
(with J. Fröhlich, J. Fuchs and I. Runkel)
in: Noncommutative Geometry and Representation Theory in Mathematical Physics;
J. Fuchs, J. Mickelsson, G. Rozenblium and A. Stolin eds., Contemporary Mathematics 391,
AMS, 2005, p. 85-100, arXiv:math/0411507v1 [math.CT]

20. “*Mini-workshop: Gerbes, Twisted K-theory and Conformal Field Theory*,” Abstracts from the mini-workshop held August 14–20, 2005. Organized by Branislav Juro, Jouko Mickelsson and Christoph Schweigert. Oberwolfach Rep. 2 (2005), no. 3, 2139–2185.
21. “*Categorification and correlation functions in conformal field theory*,”
(with J. Fuchs and I. Runkel)
Proceedings of the International Congress of Mathematicians, Madrid, 2006, p. 443-458,
arXiv:math/0602079v1 [math.CT]
22. “*Arbeitsgemeinschaft: Conformal Field Theory*,” Abstracts from the session held April 1–7, 2007.
Organized by Y. Kawahigashi, V. Ostrik and C. Schweigert.
Oberwolfach Rep. 4 (2007), no. 2, 945–994.
23. “*Defect lines, dualities and generalised orbifolds*,”
(with J. Fröhlich, J. Fuchs and I. Runkel)
Proceedings of the International Congress for Mathematical Physics, Prague, 2009.
arXiv:0909.5013v1 [math-ph]
24. “*Gerbes and Lie Groups*,”
(with K. Waldorf)
in “Trends and Developments in Lie Theory”, Progress in Mathematics 288 (2010),
Birkhäuser, K.-H. Neeb and A. Pianzola eds., p. 339-364. arXiv:0710.5467v1 [math.DG]
25. “*Bundle gerbes and surface holonomy*,”
(with J. Fuchs, Th. Nikolaus and K. Waldorf)
in: European Congress of Mathematics, Amsterdam 2008, A. Ran, H. te Riele and
J. Wiegerinck eds, p. 167 -196
26. “*Hopf algebras and Frobenius algebras in finite tensor categories*,”
(with J. Fuchs)
in: “Highlights in Lie Algebraic Methods”, A. Joseph, A. Melnikov and I. Penkov (eds.),
Progress in Mathematics Volume 295, Birkhäuser, 2012, p. 189-203
27. “*Bicategories in field theories - an invitation*,”
(with Th. Nikolaus)
in: “Strings, Gauge Fields, and the Geometry Behind”, The Legacy of Maximilian Kreuzer,
A. Rebhan, L. Katzarkov, J. Knapp, R. Rashkov and E. Scheidegger (eds.), World Scientific,
Singapore 2012, p. 119-132
28. “*The Cardy-Cartan modular invariant*,”
(with J. Fuchs and C. Stigner)
in: “Strings, Gauge Fields, and the Geometry Behind, The Legacy of Maximilian Kreuzer”,
A. Rebhan, L. Katzarkov, J. Knapp, R. Rashkov and E. Scheidegger (eds.), World Scientific,
Singapore 2012, p. 289-304
29. “*Symmetries and defects in three-dimensional topological field theory*,”
(with J. Fuchs)
in: Journal of Physics: Conference Series 597(2015) 012002.
30. “*Partition functions, mapping class groups and Drinfeld doubles*,”
(with J. Fjelstad, J. Fuchs and C. Stigner)
in: “Proceedings XXIX International Colloquium on Group-Theoretical Methods in Physics”,
C.M. Bai et al, eds., World Scientific, Singapore 2013. arXiv:1212.6916 [hep-th]

31. *“Low-dimensional topology, low-dimensional field theory and representation theory,”*
(with J. Fuchs)
in: “Representation Theory– Current Trends and Perspectives”, H. Krause, P. Littelmann, G. Malle and K.-H. Neeb (eds), EMS Series of Congress Reports, EMS Publishing House, 2017, p. 255-268
32. *“Coends in conformal field theory,”*
(with J. Fuchs)
In: “Lie Algebras, Vertex Operator Algebras, and Related Topics” K. Barron, E. Jurisich, A. Milas, K. Misra, eds, American Mathematical Society 2017, p. 65-81. arXiv:1604.01670 [math.QA]
33. *“Non-abelian gerbes and some applications in string theory,”*
(with K. Waldorf)
In: “Particles, Strings and the Early Universe : The Structure of Matter and Space-Time”, J. Haller, M. Greife eds, Hamburg, DESY 2018, ISBN: 9783945931219. arXiv:1807.05398 [hep-th]
34. *“Conformal field theories with and without boundaries: from a holographic picture to logarithmic CFT,”*
(with J. Fuchs)
In: Proceedings of the workshop “Boundary and Defect Conformal Field Theory: Open Problems and Applications,” Chicheley Hall, Buckinghamshire, UK, September 2017. arXiv:1810.05697 [hep-th]
35. *“Full Logarithmic Conformal Field theory - an Attempt at a Status Report,”*
(with J. Fuchs)
ZMP-HH/19-5, Hamburger Beiträge zur Mathematik Nr. 779, arXiv:1903.02838 [hep-th]
36. *“Module Eilenberg-Watts calculus,”*
(with J. Fuchs and G. Schaumann)
Contemporary Mathematics 771 (2021) 117-136, arXiv:2003.12514 [math.CT]

Theses:

1. *“Théories de champs conformes - aspects mathématiques et aspects physiques,”*
Habilitation à diriger des recherches. Université Paris 6, Juni 2000.
2. *“Galois and simple current symmetries in conformal field theory,”*
PhD Thesis, University of Amsterdam. June 1995.
3. *“Zur Kazama-Suzuki Konstruktion superkonformer Quantenfeldtheorien,”*
Diploma Thesis, University of Heidelberg. September 1992.

Miscellaneous:

1. *“OLAT im Übungsbetrieb von Mathematikvorlesungen,”*
(with V. Busch, S. Fleischer and M. Heinecke)
Hamburger eLMagazin 8 (2012) 14-16
2. *“Unterstützung der mathematischen Anschauung durch Visualisierungen in der Mathematik-Ausbildung für Naturwissenschaftlerinnen und Naturwissenschaftler,”*
(with R. Holtkamp)
Hamburger Universitätskolleg-Schriften 6 (2014) 82-87